NADIA KOROSTYLEVA

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Biomedical engineer with 2.5 years of experience in device development & MEMS research seeking opportunities in MedTech.

EDUCATION

M.S. & B.S. in Biomedical Engineering — M.S. GPA: **3.77 / 4.00** | Expected Graduation: **Dec. 2026**

University of South Florida (USF), Tampa, FL — B.S. GPA: **3.61 / 4.00** | Graduation: **May 2025**

SKILLS

Wet-Lab: in vivo (surgery, imaging, rodent handling); in vitro procedures on mammalian and human cell lines.

Hard Skills: 3D-printing; cleanroom & lithography; ML; embedded systems; SOLIDWORKS; Python; C++; MATLAB; COMSOL.

RESEARCH EXPERIENCE

Graduate Research Engineer | *Acousto-Bioelectronics Lab, Dr. Albert Kim – USF* — May 2025 – Present

- Lead the light therapy MEMS project **end-to-end**: designing and fabricating the **wearable**, resolving electrical and **CAD** failures to achieve a **30 °C reduction** in peak device temperature and **100× lower power consumption**; advanced from successful in vitro results, which yielded a 1st-author paper within a month, through in vivo validation (**100+ devices fabricated**).
- Leading a microfluidics chip project in collaboration with a **neuroscience lab** to extend brain slice viability 30× for long-term optogenetic studies: developed and optimized the device's microfabrication, cutting build time from 48 h to 2 h; currently advancing it through **CNC** machining, **probe integration**, and **cleanroom fabrication** to bring the platform to full functionality.
- Concurrently advancing multiple projects and papers using techniques including microfabrication with **L-edit**, **CAD design** in **SOLIDWORKS** and **Autodesk Fusion**, modeling with **COMSOL**, and data analysis with **MATLAB** and SPSS.

Research Engineer | *Gene & Regenerative Medicine Lab, Dr. Anna Bulysheva – USF* — Feb 2024 – Mar 2025

- **Operated lab solo for 6 months**: **managed** space, **quoted** supplies, **maintained** equipment, and **trained** new members.
- Designed and executed **in vitro** and **in vivo** gene electrotransfer experiments, developing and optimizing protocols across mammalian cell lines and rodent models (**IACUC certified**).
- Contributed to **multiple publications** as author and co-author, including a peer-reviewed **journal manuscript**, with full involvement in the peer-review process, and two ASGCT conference abstracts and posters.

Relevant Publications:

- **N. Korostyleva**, R. Qasim, M. Rastegar Pour, D. Yoon, J. Kim, and A. Kim, "Microneedle-mediated adaptive phototherapy (MAP) wound patch with microlens," Proc. IEEE Int. Conf. MEMS, 2026. (Conference Paper & Poster)
- J. Hensley, M. Francis, A. Otten, **N. Korostyleva**, T. Gagliardo, and A. Bulysheva, "Localized In Vivo Electro Gene therapy (LiveGT)-Mediated Skeletal Muscle Protein Factory Reprogramming," *Applied Sciences*, vol. 14, no. 23, p. 11298, Dec. 2024. (Journal Paper)

INDUSTRY PROJECTS

Artificial Homeostasis of Blood Pressure | *Senior Design – MiracleBots, LLC x USF* — Aug 2024 – May 2025

- Executed a startup-partnered project to develop a semi-automated blood pressure regulation system for ICU patients, designed in compliance with **ISO 14971, ISO 13485, ISO 81060-2,** and **IEC 60601-2-24 standards**.
- Designed and built all CAD and hardware from scratch in a team of 3 people: a non-invasive blood pressure monitoring cuff and a peristaltic IV pump for nicardipine delivery, validated through **IRB-approved** human subject testing of the cuff device.
- Oversaw and developed full software stack (**Python, C++, MATLAB**): **machine learning**-based patient response **modeling**, real-time **signal processing**, encrypted **wireless communication**, and clinician-facing application for dose control.

Wireless EOG-Based Driver Drowsiness Monitoring | *Junior Design – USF* — Apr – May 2024

- Collaborated in designing an EOG **wearable** that processes blink frequency as a clinically validated sleepiness marker.
- Developed a paired mobile app with accessible **user interface** and full **BLE communication** with the device (**C++**).
- Presented at the 2024 Florida Medical Manufacturing Consortium as **1 of the top 3 teams** in the cohort of 15 teams.

AWARDS & PROGRAMS

- NSF Research Traineeship Program in Semiconductors & Microelectronics Technology — Apr 2026 – Present
- Barbara J. Lancor BME Excellence Achievement Award: awarded to 2 out of 50 graduating BME students — May 2025
- iGEM (MIT) – Gold Medal: Awarded to Moscow 2020 team for the “CRISPR-Cas testing method for Hepatitis C” project

LEADERSHIP

Hardware Tech Lead | *IEEE Computer Society at USF* — Aug 2025 – Present

- Design and teach quarterly hands-on hardware workshops (**200+** total **attendees**, **100+ devices** built), authoring all curriculum, circuits, and firmware; topics span board fundamentals, IoT projects, **SOLIDWORKS** with **FDM/SLA** printing.
- Built **two embedded systems** and taught student how to make them: a Particle Argon optical Morse link using a reverse-biased LED as a photodetector, and an **ESP32** two-way **Wi-Fi** audio system with **I²S** MEMS microphone and Class-D amplifier.